

TABLE OF CONTENTS

M-014 INSPECTION AND TESTING

	Page No.
1. SCOPE OF WORK	M 014-2
2. CODES AND STANDARDS	M 014-2
3. DESIGN AND OPERATING CONDITION	M 014-2
4. DETAILED REQUIREMENT	M 014-2
4.1 General	M 014-2
4.2 Inspection and Tests Manufacturer's Works	M 014-3
4.2.1 General	M 014-3
4.2.2 Non-destructive Testing	M 014-4
4.2.3 Welds	M 014-4
4.2.4 Castings	M 014-5
4.2.5 Forgings	M 014-6
4.2.6 Rotor Forgings	M 014-6
4.2.7 Fabricated Components	M 014-7
4.2.8 Alloy Valves and Steam Chests	M 014-7
4.2.9 Heat Treatment	M 014-8
4.2.10 HRSG Pressure Parts	M 014-8
4.2.11 Pipework	M 014-8
4.2.12 Magnetic Particle or Dye Penetrant Inspection	M 014-9
4.2.13 Turbine, Generator and Auxiliaries	M 014-9
4.2.14 Condenser and Auxiliaries	M 014-10
4.2.15 Feed Water Heater and Heat Exchangers	M 014-11
4.2.16 Pumps-General	M 014-12
4.2.17 Boiler Feed Pumps	M 014-12
4.2.18 Valves	M 014-13
4.3 Inspection and Tests at Site	M 014-14
4.3.1 Hydrostatic Testing	M 014-14
4.3.2 Site Pipe work and Hydrostatic Test	M 014-14
4.3.3 Condenser	M 014-15
4.3.4 Turbines	M 014-15
4.3.5 Tanks	M 014-15
4.3.6 Rotating Equipment, Vibration	M 014-16
4.3.7 Rubber Lining (Piping & Tanks)	M 014-17
4.3.8 Other Tests	M 014-17

M 014 INSPECTION AND TESTING

1. SCOPE OF WORK

The Bidder shall inspect, examine and test all of the equipment and material during fabrication in shop and at site and also after erection at site.

All of the inspection, examine and test reports, shall be submitted to the Owner for review and approval.

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of **Part 4 Subsection.2.5 "Codes and Standards"**.

3. DESIGN AND OPERATING CONDITION

Not used

4. DETAILED REQUIREMENT

4.1 General

All plant and materials shall be subject to inspection and test as required for this plant. Approval of part or the whole of the plant by the Owner, or the passing of any such inspection or test, will not, however, prejudice the Owner's right to reject the equipment. The costs of all tests and inspections shall be borne by the Bidder. Furthermore, inspecting and expediting by the Owner in no way relieves the Bidder of his responsibility for quality assurance and delivery commitments.

Before any equipment is packed or dispatched from the main or Sub Bidder's Works, all relevant tests called for in this section shall have been successfully carried out in accordance with appropriate standards and to the satisfaction of the Owner or his representatives, who shall have seen and approved the test results.

The Bidder shall provide an Inspection and Test Plan for the Owner's review before finalization of the contract. At least 21 days notice shall be given before the equipment or plant is ready for shop inspection or shop test to enable the Owner to make necessary arrangements for witnessing the tests.

Every facility shall be provided by the Bidder and his Sub-Bidder for inspection or test to enable the Owner to carry out the necessary inspection and witnessing of tests.

All tests on materials and equipment shall be conducted in accordance with the codes and standards indicated in this Specifications or as mutually agreed between the Owner and the Bidder.

The Bidder shall provide test pieces as required by the Owner and shall perform mechanical tests required by the relevant codes and standards.

If any test piece fails to comply with the requirements of the appropriate Specification, the Owner reserves the right to reject the material in question and have it replaced.

The Bidder shall provide the Owner with four copies of all sub-orders and interworks orders at the time of issue. These shall be in the English Language, suitably endorsed, "Subject to inspection, and expediting by the Owner and/or their nominated representative".

Six copies of Mill and test certificates, which shall be in English, shall be supplied to the Owner. The Bidder shall supply at the ex-works stage six bound copies of all certification, suitably indexed to cover all tests of material and equipment.

Where type test are specified, in addition to routine tests the Bidder may submit to the Owner type test certificates in lieu of actual type tests, provided these have been independently witnessed and cover identical items of plant.

For all non destructive examination (NDE), the Bidder shall propose the best of Acceptance Criteria Levels to be used for testing subject to Owner's approval, unless otherwise specified.

4.2 Inspection and Tests Manufacturer's Works

4.2.1 General

Works inspection shall include mechanical and hydraulic tests, as well, as running and functional tests (under simulated conditions) on the assembled equipment, prior to shipment to the Site, in accordance with the relevant standards or as otherwise agreed with the Owner.

4.2.2 Non-destructive Testing

Materials requiring non-destructive tests, except where stated otherwise in this Specification, shall be performed in accordance with the following:

- ASTM E114 Standard Practice for Ultrasonic Pulse-Echo Straight-Beam Examination by the Contact Method
- ASTM E94 Standard Guide for Radiographic testing
- ASTM E142 Controlling Quality of Radiographic Testing.
- ASTM E709 Standard Guide for Magnetic particle inspection
- ASTM E165 Standard Test Method for Liquid penetrant inspection

4.2.3 Welds

Prior to commencement of any welding of the work, the Bidder shall have all welding procedures, welding equipment and operators qualified in accordance with ASME Boiler code Section IX welding qualifications or other standards as approved by the Owner. All welding procedure with exception of propriatery ones shall be submitted to the Owner for his approval. A list of the qualified operators to be used in the work must be available for the Owner's inspection prior to start of any welding.

Strict measures of quality control shall be exercised throughout the work.

The Owner may call for a radiograph of the work of any operator who in his opinion is not maintaining the standard of workmanship required. Where the interpretation of radiograph proves the welding to be of poor quality, the Owner reserves the right to re-examine all radiographs on all production welds done by the operator. Should the welding prove unacceptable, the operator is to be removed from any further production welding until requalified, and all defective or sub-par weld shall be re-worked by other qualified operator. Indonesian welders who are involved in this project shall have certificates issued by Man Power Department of Government of Indonesia, or an authorized Third Party Inspection Company.

Ultrasonic inspection when called for in the code or in this Specification shall be in accordance with:

- ASTM E164 Standard Practice Ultrasonic contact Examination of Weldments

Imperfection found by ultrasonic examination shall be judged against the acceptance - rejection standards outlined in ASME Code Section VIII Div. 1, Appendix 12, and in the case of radiographic examination as outlined in paragraph UW 51 or UW 52 as applicable.

4.2.4 Castings

Each pressure retaining casting and each major load-bearing casting shall have an integral test bar, wherever possible, from which mechanical test specimens can be taken. Test specimens shall be in accordance with ASTM A370 "Methods and Definitions for Mechanical Testing of Steel Products" and ASTM E8 "Tension Testing of Metallic Materials". Non-destructive testing when required shall be carried out in accordance with the standards outlined in Clause 4.2.2. The standard as mentioned shall be the latest edition.

Defects in castings which have been found by radiography, magnetic particle or liquid penetrant inspection are to be assessed against the following standards which shall be used as a basis for acceptance or rejection of the casting.

- ASTM E186 Ref. radiographs for heavy walled steel casting 51 mm - 144 mm thick
- ASTM E280 Ref. radiographs for heavy walled steel casting 144 mm - 305 mm thick
- ASTM E446 Standard Ref. radiographs for heavy walled steel casting up to 51 mm thick
- ASTM E12 Ref. radiographs for magnetic particle indications on ferrous castings

For N.D.T acceptance/rejection standards refer to ASME Sect. VIII Div. 1.

Except where stated otherwise the limit of acceptability for all major castings such as boiler and turbine stop valves, turbine cylinders, strainer chests and boiler feed pump casings which have been radiographed, shall be severity level 1. All parts of castings over 300 mm thick shall be ultrasonically examined.

Castings rejected through the application of the above techniques may be repaired by welding, subject to meeting the requirements states in ASME Section VIII, Division 1 and ASTM A-488 "Recommended Practice Qualification of

Procedures and Personnel for the Welding of Steel Castings". Repairs will not be allowed to grey iron castings having injurious defects.

Where a casting forms part of welded assembly, the welds shall be examined in accordance with ASME Section VIII, Division 1, paragraph UW 51.

Boiler feed pump castings, impellers and diffusers shall be subjected to a complete magnetic particle or dye penetrant test. In addition radiographs shall be taken on all impellers, covering the shroud areas, from the neck to the outside diameter.

4.2.5 Forgings

Test pieces shall be provided for all steel forgings and subjected to physical tests in accordance with ASTM A370 and ASTM E38, except where specified otherwise in this specification. All steel forgings after rough machining shall be subject to an ultrasonic examination in accordance with ASTM A388 "Ultrasonic Examination of Heavy Steel Forgings". Magnetic particle inspection shall also be carried out at areas of fillets and change of section in accordance with ASTM A275 "Magnetic Particle Examination of Steel Forgings".

Imperfection found by ultrasonic examination shall be judged against the acceptance/rejection standards outlined in ASME Section VIII.

4.2.6 Rotor Forgings

Rotor for gas turbines, steam turbines, generators, boiler feed pumps and other equipments, shall be tested and inspected by Ultrasonics in accordance with ASTM A418 "Ultrasonic Inspection of Turbine and Generator Steel Forgings".

The choice of rotating or stationary tests shall be based on the specific design of the rotor to be tested and shall be subject to the approval of the Owner. If required the Bidder shall perform thermal stability tests on turbine rotors in accordance with ASTM A472.

All generator rotor forgings shall be subject to fracture appearance transition temperature tests, and magnetic permeability and resistivity tests. An ultrasonic examination shall be carried out in accordance with ASTM A418 after rough machining.

Ultrasonic and magnetic permeability tests shall be carried out for rotor end-winding retaining ring forgings.

The retaining ring shall be ultrasonic and dye penetrant checked in the rough machined state and hydraulically tested followed by dye penetrant inspection after final machining. Ultrasonic inspection shall be in accordance with ASTM A351 "Ultrasonic Inspection of Turbine Generator Steel Retaining Rings".

4.2.7 Fabricated Components

All components machined or fabricated from plate, sheet or bar stock shall meet the material requirements of ASTM or material specification approved by the Owner.

All components, shall be subjected to two or more of the following tests, as agreed by the Owner:

- Visual
- Dye penetrant
- Magnetic particle
- Ultrasonic
- Radiographic

These tests shall be in accordance with the recommended practices of the ASTM. The terms of reference for acceptance shall be the applicable ASTM specifications or equivalent standard as applicable.

4.2.8 Alloy Valves and Steam Chests

Alloy steel valves, steam chests and fittings shall be supplied complete with forged stub ends of material suitable for site welding of the adjoining pipe. Weld preparations shall be subjected to a crack detection test and be fully radiographed for a distance of 75 mm from each end in accordance with ASTM E94 "Recommended practice for Radiograph Testing" and shall be judged by reference Radiographs for Steel Castings ASTM E186 and E446 as applicable. Acceptance standards for crack detection tests shall be in accordance with ASTM Section VIII Division 1.

4.2.9 Heat Treatment

All parts subject to heat treatment shall have adequate records kept, giving details of rate of rise and fall of temperature, holding time, temperature and method of cooling.

4.2.10 HRSG Pressure Parts

All HRSG pressure parts shall be tested in accordance with the requirements of the ASME Boiler and Pressure Vessel Code unless otherwise specified.

4.2.11 Pipework

All fabricated pipework shall be in accordance with standards as specified by ANSI B31.1 "Power Piping Code", latest edition.

Temperature records of hot forming and heat treatment operations shall be made, and submitted to the Owner.

The wall thickness of hot formed bends shall be checked by ultrasonic method. Records of these tests shall be kept and made available to the Owner.

Where random radiography is specified, it shall refer to the final weld only, and shall be at Owner's selection and discretion.

For the purposes of radiographic inspection a root pass is defined as the first complete weld run.

All piping not covered by 100% radiography shall be hydraulically tested to 1.5 times the design pressure.

The following table gives the butt weld radiographic requirements for the major steam, water, oil and gas lines.

In addition to the NDT requirements stated in ANSI B31.1 para 136.

Percentage given are percentages of the total number of welds to be examined by radiographic testing for nominal pipe sizes over 50 mm and by magnetic particle or liquid penetrant testing for sizes 50 mm and less.

Table 1: Quantitative Non-destructive Examination Requirements for Piping Systems.

Piping	Root Pass	Final Pass (After PWHT where applicable)
Main Steam (both HP and LP), HP auxiliary and gland steam	100%	100%
HP Turbine by-pass, loop pipes	100%	100%
L.P Auxiliary steam, make-up and recirculation drain system	None	10%
All drain system, and vents	None	10%
HP feedwater	100%	100%
LP bypass, LP feedwater, condensate, oil system, air system	None	20%
Fuel oil, natural Gas, bleed steam, all process water (including circulating and closed cooling) system, CO2 System	None	10%

The above requirements are applicable to both site and shop welds.

4.2.12 **Magnetic Particle or Dye Penetrant Inspection**

All nozzles, branch connections, butt welds and attachment welds in alloy steel pipe shall be subjected to magnetic particle or dye penetrant inspection.

Magnetic particle inspection shall be carried out in accordance with ASTM E709.

When dye penetrant method is used, it shall be conducted in accordance with ASTM E165 and applied as directed by the penetrant manufacturer.

4.2.13 **Turbine, Generator and Auxiliaries**

The Bidder shall perform all tests and inspection necessary to ensure that the material and workmanship conform to the Contract and design drawings and that such tests are adequate to demonstrate that the equipment will comply with the requirements of this Specifications and meet the guarantees specified.

The generator and the turbine rotors shall be overspeed tested at 20% above rated speed for two minutes and balanced in accordance with IEC Publication 45. The generator rotor test shall be carried out to demonstrate the absence of shorted turns in the rotor winding: These tests shall be performed throughout the speed range of the generator. High voltage dielectric tests and insulation resistance measurements shall be carried out on the rotor winding before and after the overspeed test.

All items such as assemblies, sub-assemblies and systems shall be subject to inspection by the Owner. Such items shall include, but not be limited to: casing, bedplate, coupling, coolers, bearings, stator core, rotor, brushgear, terminations, auxiliaries, piping. Systems such as lubrication, seal oil, demineralised water and shall have both their individual components and the whole system functionally tested.

Where possible functional tests of the systems shall be carried out connected to the generator, and before shipping to Site.

Where the generator or main parts are not fully assembled in the shop, the items shall be trial fitted together before shipping.

The build-up tolerances shall be checked and recorded.

The generator stator casing and endshield shall be hydrostatically pressure tested, for 15 minutes to specified design pressure.

A ring flux test shall be carried out on the completed stator core to demonstrate absence of local hot spots.

All parts subject to steam or water pressure shall be tested with hydraulic pressure of 50% in excess of the highest working gauge pressure to which the component will be subjected.

All equipment shall be tested under simulated working conditions and conducted in accordance with the appropriate ASME Power Test Code.

4.2.14 Condenser and Auxiliaries

The entire condenser, including waterbox, covers, access doors, tube sheets, tube support plates, baffles, waterbox expansion elements, steel rod dummy

tubes and all stay bars shall be assembled in the Bidder's or Sub-Bidder works to ensure correct alignment of mating pieces and the supports.

All condenser tubes shall be pneumatically tested to 10 bar(g) internal pressure and carefully inspected prior to installation.

Should the Owner deem it necessary, the Bidder shall perform an ultrasonic test on the condenser tubes in addition to the hydraulic test. The air ejectors and quick start exhausters shall be tested in accordance with ASME Power Test Code PTC 24 for Ejectors and Boosters.

4.2.15 Feed Water Heater and Heat Exchangers

All materials, welding and non-destructive testing shall meet the requirements outlined in ASME Section VIII "Pressure Vessel Code" and ASME Section IX "Welding Qualification".

All parts subject to steam or water pressure shall be tested at a hydraulic pressure according to the design code.

All closed cooling water heat exchanger tubes shall be tested according to the design code requirement. Circulating Water and Closed Cooling Water System & Screen

The CW pumps shall be tested at the specified minimum submergence and static lift to confirm pump total head and power requirements between shut-off, maximum rated flow and run-out conditions. On completion of the tests the submergence shall be decreased to verify the NPSH requirements at maximum rated flow.

The CCW pumps shall be tested to determine head/capacity, and head/power characteristic curves and to verify freedom from vibration between shut-off, maximum rated flow and run-out conditions.

Each traveling band screen shall be completely assembled in the Bidder's or Sub-Bidder works to ensure correct alignment of mating parts and supports.

All parts subject to pressure shall be hydraulically tested according to the design code.

4.2.16 Pumps-General

All casting and forgings shall meet the requirement stated in Clause 4.2.4 and 4.2.5.

All parts subject to pressure shall be tested according to the design code requirement.

Pumps testing shall be in accordance to requirements in Part 4 Section 2.6.

Certified pump performance curves and list certificates shall be submitted to the Owner for approval, and for records.

Pumps shall not be shipped until the Owner has seen and approved the test results.

Where applicable, pumps shall be assembled with their spare inner assemblies and shall be shown to be interchangeable.

4.2.17 Boiler Feed Pumps

Impellers shall be dynamically balanced.

The pump rotating element complete with the pump-half coupling, balancing drum, and thrust bearing collar mounted on the pump shaft shall be dynamically balanced and checked for concentricity.

Each HRSG feed pump, flexible coupling, drive motor and lubricating oil system(s) shall be assembled in the manufacturer's works and the required works tests carried out.

A performance test shall be carried out on each combined pump and motor assembly at the Bidder's works to demonstrate that the pump complies fully with the Specification.

Prior to the conducting of the performance test, the Bidder shall submit a performance test procedure including test set-up description for the Owner's approval.

On completion of the performance tests the manufacturer shall prepare curves to illustrate that there are no deviations from the design characteristics.

While under test, the pumps shall be checked for shaft vibration at each bearing.

Unless otherwise stated or agreed by the Owner each rotating machine has to comply with the requirements for vibration magnitude (criterion I) designation as “zone A” stipulated by ISO Specification 10816 for the respective group of machinery. If the vibration magnitude is greater than stipulated as “zone B”, the Owner has the right to reject the concerned equipment.

The first critical speed shall be determined. The pumps shall perform with no undue heating in the bearings or other evidence of unsatisfactory design or construction such as noisy impellers or undue gland leakage. Spot suppression tests shall be carried out to verify pump NPSH requirements.

If the test results indicate necessity for trimming impellers or other modification, details of such work shall be submitted to the Owner for review before proceeding with the work.

4.2.18 Valves

- a. Valves
- b. Valves shall be hydrostatically tested in accordance with the Valve Manufacturers Standardization Society Standard MSS-SP-61 "Hydrostatic Testing of Steel Valves" Circulating Water Butterfly Valve

The Bidder shall test the butterfly valves at his works prior to shipping, in accordance with AWWA Standard 0504.

- c. Power Operated Valves

Completed valves and actuators shall be tested under load to determine the following:

- Valve operating time to open and close.
- Clutch coil current at rated voltage.
- Motor currents at rated voltage for the following condition:

Starting

Full load

Maximum during opening operation

Maximum during closing operation

4.3 Inspection and Tests at Site

The Bidder shall submit to the Owner for approval a comprehensive quality control plan, covering the pressure parts of the HRSG, erection of the turbine generator, welding of the associated pipework, and erection of all ancillary plant and the electrical equipment.

This document shall identify the number of supervisory personnel and degree of inspection and tests expected to be carried out during erection up to completion. It should be equal to that level of control which has been applied at the Works and in compliance with the codes and standards identified in this Specification. In addition, records of all welders' qualification are to be submitted to the Owner for approval prior to welding. Records of all heat treatment and non-destructive tests carried out are also to be submitted to the Owner.

4.3.1 Hydrostatic Testing

Heat Recovery Steam Generator

The Bidder shall, upon completion of erection, subject all the pressure parts to a hydraulic pressure test which meets ASME Boiler and Pressure Vessel

Code requirements. All pressure parts shall be subjected to the test prior to any boiler casing installation. The Bidder shall supply and install all equipment, piping and materials, including the safety valve gauges and pumps, required and necessary for the tests.

4.3.2 Site Pipe work and Hydrostatic Test

The table for butt weld radiography given in this Section shall be used as the basis for the extent of radiography requirements at Site.

Where radiographs of Site butt welds joint root or initial passes are called for, they shall be radiographed hot by the Bidder, subject to the minimum weld deposit thickness as given in ANSI Power Piping Code, B.31.1, Section 131.

The Bidder shall perform hydrostatic test on all pipework not covered by 100% radiography to a pressure not less than 1.5 times design pressure. These tests shall be carried out in the presence of the Owner or his representatives. For the

purposes of the hydrostatic test, the Bidder shall provide temporary blanking plates, as required.

4.3.3 Condenser

Prior to tube installation, a local test shall be performed to detect any possible tube imperfections such that no tube plugging will be required upon completion of the tightness test described below.

Tightness test shall be conducted to ensure that no leakage occurs between the steam and the circulating water sides of the condenser.

The test shall be conducted by flooding the steam side of the condenser with water, detergent, and an additive sensitive to ultra violet light for inspection of leakage at the tube sheets. It shall not be necessary to provide additional support of the condenser bottom during this test.

The water side of the condenser shall be hydrostatically tested to specified pressure measured at the top of the waterbox.

4.3.4 Turbines

The following tests shall be conducted by the Bidder after installation of the equipment or system concerned:

- Functional tests to demonstrate that the controls are operating as intended.
- Tests to demonstrate that the governor system deadband is within specified limits.
- Dynamic response tests for step error signals at selected loadings.
- Capacity tests to determine adequacy of the hydraulic system to supply requirements of actuators, etc.
- Additional tests as may be required to demonstrate the governing system's performance when the unit is connected to the Owner's transmission system.
- Functional Testing of all equipment.

4.3.5 Tanks

- a. Vertical Tanks

All tanks constructed at site, except where specified otherwise in this Specification, shall be subject to inspection and testing, in accordance with the requirements stated in API 650.

The Bidder before carrying out any welding shall meet the conditions stated in Clause 4.2.3 of this Specification.

Checking of the welded seams shall be by spot radiography. The number and location of radiographs shall be as specified in API 650, Section 6. The Owner reserves the right to call for additional non-destructive tests as he may consider necessary.

On completion, all tanks shall be subjected to the tests prescribed in API 650, Section 5.

b. Horizontal Tanks

All tanks constructed at site, except where specified otherwise in this

Specification shall be subject to inspection and testing, in accordance with the requirements stated in ANSI/AWWA D100 Standard for Welded Steel Tanks for Water Storage.

The Bidder before carrying out any welding shall meet the conditions stated in Clause 4.2.3 of this Specification.

Checking of the welded seams shall be by spot radiography except those welds excluded by ANSI/AWWA D100. The number and location of radiographs shall be as specified in ANSI/AWWA D100 Section 11.

On completion, all tanks shall be subjected to the tests prescribed in ANSI/AWWA D100 Section 11 "Testing".

4.3.6 Rotating Equipment, Vibration

Rotating Equipment such as pumps shall be subject to a test for checking correct rotation, and to verify vibration and noise guarantees.

Unless otherwise stated or agreed by the Owner each rotating machine has to comply with the requirements for vibration magnitude (criterion I) designation

as “zone A” stipulated by ISO Specification 10816 for the respective group of machinery. If the vibration magnitude is greater than stipulated as “zone B”, refer to legal section Part 3, Special Conditions of Contract, Clause 4.2.1.

Noise emissions shall meet the requirements of **Part 4 Section 2.9 and 2.16**.

4.3.7 Rubber Lining (Piping & Tanks)

Pin-hole test and visual check of complete lining including any patched portion shall be carried out. The voltage used shall be 75 % of the breakdown voltage of the coating. This break-down voltage shall be first determined using test plates coated with the specified rubber lining and thickness, separately. The coating on the test plate shall also be micro-sectioned, by the applicator to show that it is free from vacuoles and other defects.

If the defects revealed by the above test procedure do not exceed one part per 5 m² of coating surface, the coating need not be retested after the defects have been repaired. If the defects exceed one part, per 5 m² of coating surface, the repairs shall be retested, and this procedure shall be repeated until the defects are less than one part, per 5 m² of coating surface.

4.3.8 Other Tests

In addition to the tests listed above, the Bidder shall demonstrate the function and capability of all equipment and systems supplied under the Contract, to perform in accordance with the Specification requirement, and shall be conducted in accordance with the testing procedures.

(Last page of M-014)